



ST BENEDICT'S

SUBJECT

Mathematics

PAPER

**Analytical
Geometry and
Statistics**

GRADE

11

DATE

8 April 2019

EXAMINER

Mr. Benecke

MARKS

55

NAME

MODERATOR

Grade 11 Teachers

TEACHER

DURATION

1 hour

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Analytical Geometry (Part 1)	17	
2	Analytical Geometry (Part 2)	12	
3	Box-and-Whisker	13	
4	Linear Regression	7	
5	Ogive	6	
TOTAL		55	

INSTRUCTIONS:

1. This paper consists of 5 questions and 10 pages.
2. Read the questions carefully.
3. Answer all questions.
4. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
5. Round off your answers to two decimal digits where necessary.
6. All necessary working details must be shown. Answers only, without the relevant calculations will not be given marks. Equations may not be solved solely with a calculator.
7. It is in your interest to write legibly and present your work neatly.
8. Formulas you may need are listed below:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2} ; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$\bar{x} = \frac{\sum fx}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

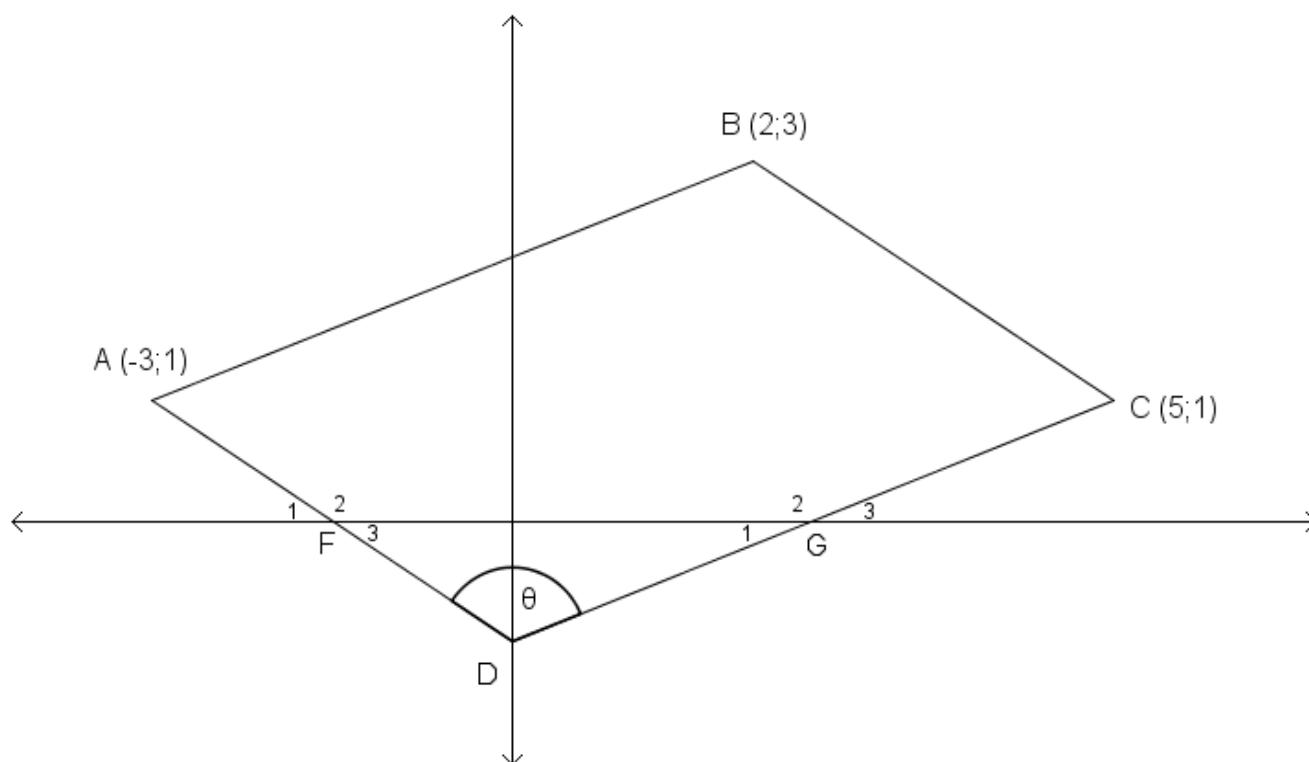
$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

QUESTION 1

17 MARKS

In the diagram below, ABCD is a parallelogram given points A (3;1), B(2;3), and C(5;1).



a) Determine the length of line AB.

(2)

$$AB = \sqrt{(2+3)^2 + (3-1)^2} \quad \checkmark$$

$$AB = \sqrt{25+4}$$

$$AB = \sqrt{29} \quad \checkmark$$

b) Determine the gradient of line AB.

(1)

$$m = \frac{3-1}{2+3}$$

$$m = \frac{2}{5} \quad \checkmark$$

c) Determine the equation of line CD. (3)

$$y = \frac{2}{5}x + c \checkmark$$

$$1 = \frac{2}{5}(5) + c \checkmark$$

$$c = -1$$

$$y = \frac{2}{5}x - 1 \checkmark$$

d) Find the coordinates of point D. (2)

$$y = \frac{2}{5}x - 1$$

$$y = \frac{2}{5}(0) - 1 \checkmark$$

$$y = -1$$

$$D (0; -1) \checkmark$$

e) Determine the coordinates of point M, the intersection of the diagonals of $ABCD$. (2)

$$M (x; y) = M \left(\frac{-3+5}{2}; \frac{1+1}{2} \right) \checkmark$$

$$= M(1; 1) \checkmark$$

f) Calculate the value of angle $\hat{G}_3$. (2)

$$\tan \hat{G}_3 = \frac{2}{5} \quad \checkmark$$

$$\hat{G}_3 = 21.80^\circ \quad \checkmark$$

g) Calculate the value of θ if $D(0; -1)$. (5)

$$\hat{G}_1 = 21.80^\circ$$

$$\tan \hat{F}_3 = \text{grad } AD$$

$$\tan \hat{F}_2 = -\frac{2}{3} \quad \checkmark$$

$$\hat{F}_2 = 180 - 33.69^\circ$$

$$\hat{F}_2 = 146.31 \quad \checkmark$$

$$\theta = 146.31^\circ - 21.80^\circ \quad \checkmark \quad (\text{Ext angle of triangle}) \quad \checkmark$$

$$\theta = 124.51^\circ \quad \checkmark$$

QUESTION 2

12 MARKS

Given $P(3;5)$; $Q(k;6)$; $R(-3,7)$ and $S(-2;-2)$.

Determine the value of k if:

a) $PQ \perp RS$. (4)

$$\frac{6-5}{k-3} \times \frac{7+2}{-3+2} \checkmark = -1 \checkmark$$

$$\frac{1}{k-3} \times \frac{9}{-1} = -1 \checkmark$$

$$\frac{9}{-k+3} = -1$$

$$9 = k - 3$$

$$k = 12 \checkmark$$

b) $QR = QS$. (4)

$$(k+3)^2 + (7-6)^2 \checkmark = (k+2)^2 + (6+2)^2 \checkmark$$

$$k^2 + 6k + 9 + 1 = k^2 + 4k + 4 + 64 \checkmark$$

$$2k = 58$$

$$k = 29 \checkmark$$

c) QRS is collinear

(4)

Gradient of QR = gradient of RS

$$\frac{7-6}{-3-k} \checkmark = \frac{-2-7}{-2+3} \checkmark$$

$$\frac{1}{-3-k} = \frac{-9}{1} \checkmark$$

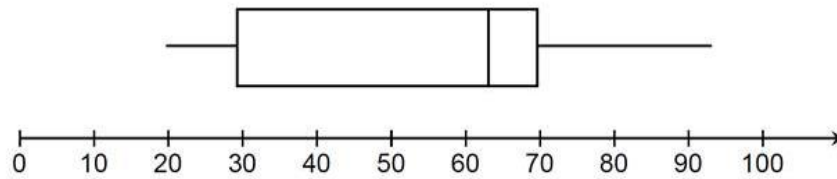
$$27 + 9k = 1$$

$$k = -\frac{26}{9} \checkmark$$

QUESTION 3

13 MARKS

- a) Consider the following box-and whisker diagram. It represents the results of a Grade 12 mathematics test.



- 1) Write down the five number summary. (5)

Min: 20 ✓

Q_1 : 30 ✓

Med: 63 ✓

Q_3 : 70 ✓

Max: 92 ✓

- 2) Determine the inter-quartile range. (1)

IQR: 40 ✓

- 3) Comment on the skewness and explain what it means in the given situation. (2)

Negatively skewed/skewed to the left. ✓ Marks spread out on the left/marks closely grouped together on the right. ✓

- b) Three numbers 6, 7, and p have a mean of 8.

- 1) Determine p . (1)

$$\frac{6+7+p}{3} = 8$$

$$p = 11 \text{ ✓}$$

- 2) Determine the standard deviation without using a calculator. Show all working out. (4)

$$\sigma = \sqrt{\frac{(6-8)^2 + (7-8)^2 + (11-8)^2}{3}} \text{ ✓ ✓}$$

$$\sigma = \frac{\sqrt{42}}{3} \text{ ✓}$$

QUESTION 4**7 MARKS**

The number of hours studied, x , and the grade (in percent) achieved, y , are measured for 4 learners at St. Benedicts.

x	12	13	12	15
y	62	70	72	85

- a) Determine the equation of the line of best fit. (2)

$$y = 6x - 5,75$$

- b) Calculate the correlation co-efficient. (1)

$$r = 0,89$$

- c) What grade would you estimate a learner achieving if he spending 10 hours studying? (1)

$$y = 6(10) - 5,75$$

$$y = 54,25$$

- d) What grade would you estimate a learner achieving if he spending 8 hours studying? (1)

$$y = 6(8) - 5,75$$

$$y = 42,25$$

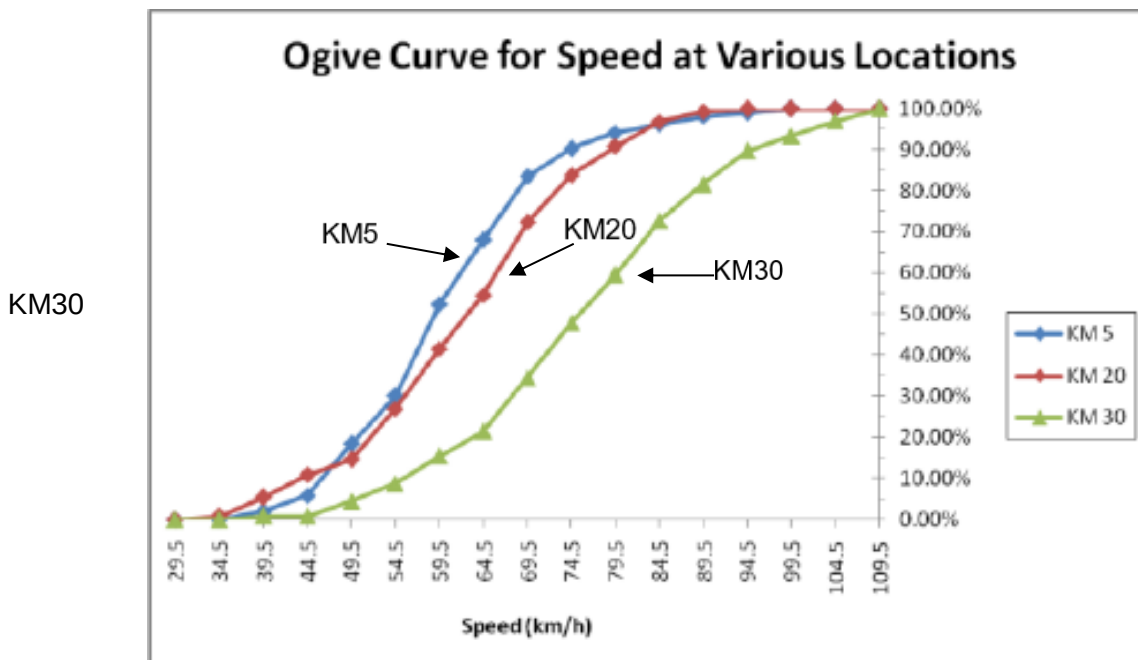
- e) Is the estimate in (c) or (d) more reliable? Give a reason for your answer. (2)

(c) 10 hours studying is more reliable, closer to the data set.

QUESTION 5

6 MARKS

The ogive (cumulative frequency curve) below represents speed distribution on 3 different roads in Malaysia. The roads are called KM5, KM20, and KM30.



a) State whether the following statements are true or false. If false, correct the statement. (4)

1) The highest speeds were recorded on KM30.

True ✓

2) The 85th percentile speed is 69,4km/h, 74,5km/h and 89,6 km/h respectively.

True ✓

3) The median speed on KM20 is approximately 69,5km/h.

False ✓, The median speed on KM20 is approximately 64,5km/h. ✓

b) Complete the following statements:

1) 10 ✓ _____ % of drivers on KM5 exceeded 75km/h. (1)

2) The speed limit on KM30 is 90km/h. 80 ✓ _____ % of drivers adhered to this. (1)